Korea Technology and Information Promotion Agency for SMEs

*(max 10 pages)*

# Technology Proposal for 2026 TIPA* International Joint R&D Program

**(TIPA: Korea Technology and Information Promotion Agency for SMEs)**

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

## 1. Technology Proposer Information

| Proposer | Name | Paul Mentink | Email | paul.mentink@tno.nl |
|---|---|---|---|---|
| Institute Name | TNO (Netherlands Organisation for Applied Scientific Research)<br>Unit Mobility & Built Environment, Powertrains Department | | | |
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## 2. Technology Overview

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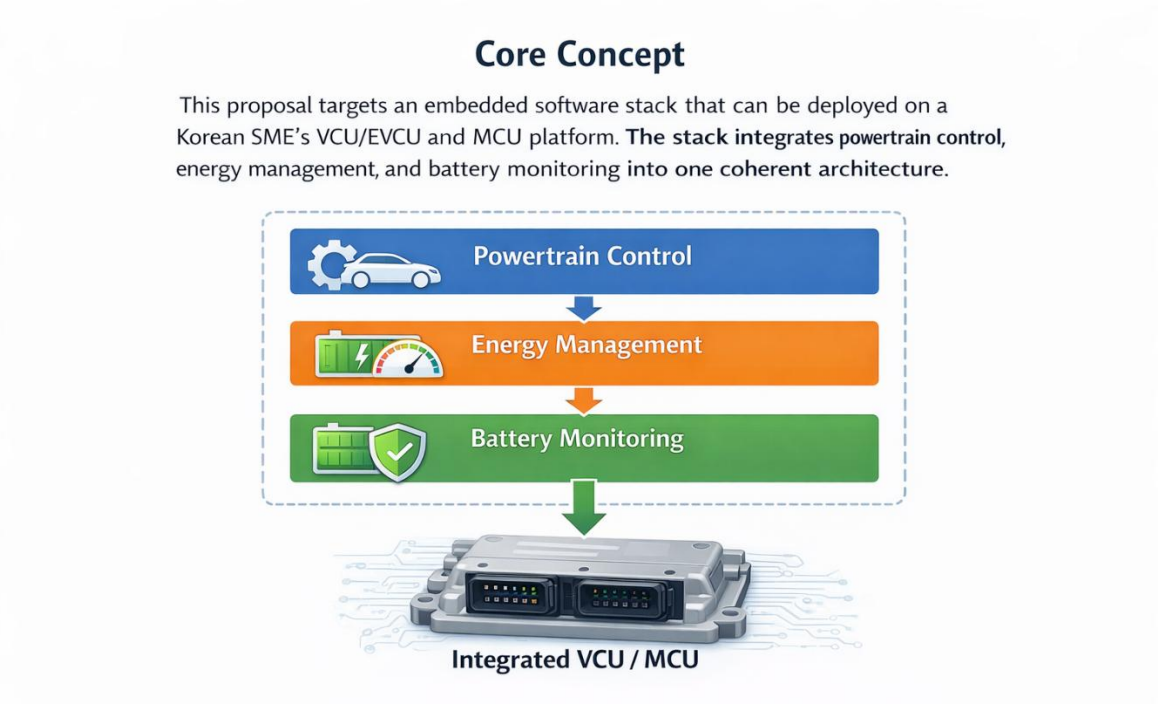
|---|---|---|---|---|
| Title of Proposed Technology | Integrated VCU-based powertrain and battery health management software (physics-informed modelling, modular energy management, and anomaly diagnostics for EV and hybrid applications) | | | |
| Korea's 12 National Strategic Technology Fields | Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( ) | | | |
| Industrial Technology Classification [1] | **Technology Field 1** (Large Category) | | **Technology Field 2** (Large Category) | |
| | Electrical & Electronics | | Electrical & Electronics | |
| | Medium Category | Small Category | Medium Category | Small Category |
| | Batteries | Secondary Batteries | Heavy Electrical Equipment | Automated control equipment |
| Concept of Proposed Technology | Core concept<br>This proposal targets an embedded software stack that can be deployed on a Korean SME's VCU/EVCU and MCU platform. The stack integrates powertrain control, energy management, and battery monitoring into one coherent architecture. | | | |




Technical building blocks (TNO contribution)

- Modular Energy Management (MEM): supervisory energy management strategies designed for efficient deployment on vehicle control units, as described by TNO (Mentink) [R1].

- Physics-based modelling and real-time implementation: TNO Powertrains states that its component-library based models are physics-based, validated in TNO facilities, fully predictive and can be implemented in real-time [R2].

- Battery modelling and BMS toolchain: TNO describes a model-based development workflow and BMS algorithm development toolchain using MATLAB/Simulink and in-house tools, including a Battery Modular Simulation Tool (BMST) [R7].

- Powertrain simulation environment: TNO developed ADVANCE as a modular simulation tool for design and evaluation of advanced powertrains [R3].

- Battery digital twin and optimized charging algorithms: TNO publications include optimized fleet charging with quantified cost savings and real-time feasible computation [R4], and charge scheduling methods that use machine learning for remaining useful life aware decisions [R5]. Related work on battery digital twin-based charge profile prediction is also available via the TNO repository [R6].

Korean SME contribution

- VCU/EVCU and MCU hardware and software platform, including integration in EV and hybrid powertrains (scope based on internal TIPA SME list) .

- Access to representative powertrain requirements, calibration data, and demonstrator

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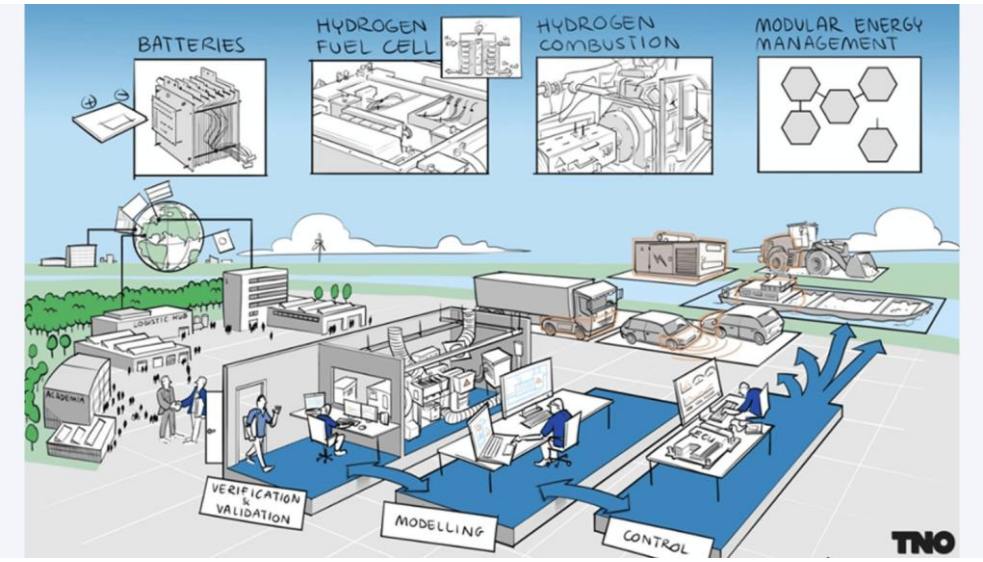
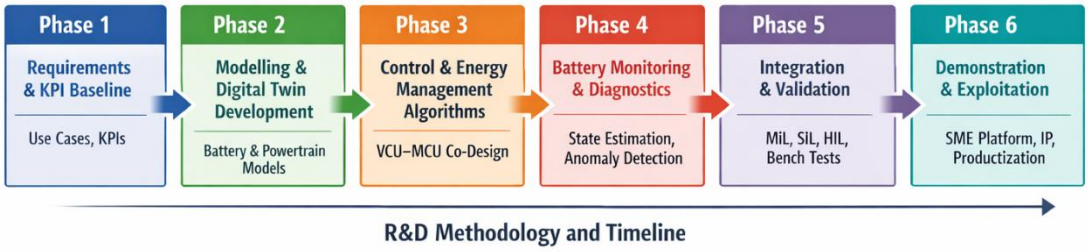
| | integration support. |
| **Rationale for Joint R&D Project** | Need for joint R&D and rationale for collaboration<br><br>- The Korean SME should have product focus on VCU/EVCU and MCU development and powertrain integration. This provides a concrete route to productization for embedded algorithms.<br><br>  <br><br><br>- TNO contributes validated modelling, control, and testing capabilities, including battery monitoring and control expertise and dedicated test facilities for batteries and powertrains (TNO Powertrains Test Centre) [R2].<br><br>- Collaboration is required to: (i) co-design algorithms with the SME's controller constraints, (ii) integrate into the SME software stack and validate in representative system environments, and (iii) jointly develop IP and demonstrators that can be scaled to commercial EV and hybrid platforms. |
| **Key Words** | 1. Vehicle Control Unit (VCU/EVCU)<br><br>2. Motor Control Unit (MCU)<br><br>3. Modular Energy Management (MEM)<br><br>4. Battery Management System (BMS)<br><br>5. Physics-informed diagnostics and anomaly detection |

## 3. Research and Development Objectives & Plan

| Technology Readiness Level (TRL)[2] | Start of the Project | Level (4) | End of the Project | Level (6) |
|---|---|---|---|---|
| **R&D Objectives** | R&D Objectives<br><br>1) Develop an integrated control and monitoring architecture for EV and hybrid powertrains that combines VCU/EVCU supervisory control, MCU coordination, and battery monitoring functions.<br><br>2) Implement Modular Energy Management (MEM) supervisory control functions that can be executed on embedded vehicle control units, building on TNO's MEMs approach [R1].<br><br>3) Develop model-based battery state estimation and health diagnostic functions supported by physics-based models and validated toolchains at TNO [R2][R7].<br><br>4) Develop battery and powertrain anomaly detection functions that fuse model-based residuals with data-driven methods, with clear diagnostic outputs suitable for on-board and off-board use.<br><br>5) Validate the complete software stack through simulation (ADVANCE), hardware-in-the-loop, and bench testing using TNO Powertrains Test Centre facilities where relevant [R2][R3]. | | | |
| **R&D Target Specifications** | R&D Target Specifications (project targets)<br><br>- Embedded execution: Real-time capability on SME controller hardware with control cycle time down to 10 ms and deterministic execution (target), with documented CPU and memory footprint.<br><br>- Energy management benefit: At least 5% reduction in energy consumption for selected reference duty cycles versus SME baseline strategy (target).<br><br>- Battery state estimation: SoC RMS error below 2% and SoH estimation error below 5% on representative datasets (target).<br><br>- Diagnostics: Detect selected failure modes (sensor faults, actuator faults, abnormal battery behavior) with false positive rate below 1% and detection time below 60 s in defined test scenarios (target).<br><br>- Demonstrator maturity: Integrated prototype validated at system level (target TRL 6 at project end). | | | |
| **R&D Plan** | Draft R&D Plan (36 Month indicative) | | | |




**Phase 1: Requirements and baseline definition (Month 1–4)**

- Define functional and safety-relevant use cases for integrated VCU/MCU and battery monitoring.

- Establish baseline controller and monitoring performance and agree on KPI definitions.

**Phase 2: Modelling and digital twin development (Month 3–10)**

- Develop or adapt physics-based battery and powertrain models with parameter identification workflows (TNO modelling approach) [R2][R7].

- Create digital twin interfaces for charge and energy prediction, building on TNO work on charge profile prediction and charge scheduling [R5][R6].

**Phase 3: Control and energy management algorithm development (Month 8–18)**

- Implement MEM supervisory control strategies that coordinate torque request, power limits, and thermal constraints, tailored to the SME VCU constraints [R1].

- Co-design of control interfaces between VCU supervisory layer and MCU low-level control.

**Phase 4: Battery monitoring and diagnostics (Month 12–22)**

- State estimation using observer based methods (for example EKF or UKF) and parameter tracking to capture aging.

- Diagnostics and anomaly detection combining model residuals, plausibility checks, and data-driven detectors.

**Phase 5: Integration and validation (Month 20–32)**

- Model-in-the-loop and software-in-the-loop validation using TNO toolchains (including ADVANCE for powertrain simulations) [R3].

- Hardware-in-the-loop and bench validation. TNO Powertrains Test Centre provides climate chambers and electrical test benches for batteries and powertrains, supporting validation and certification activities [R2].

**Phase 6: Demonstration and exploitation (Month 30–36)**

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| | - Integrated demonstrator on SME platform in representative environment.<br>- Joint IP strategy, documentation, and productization plan for integration into SME controller portfolio. |

## 4. R&R (Roles and Responsibilities)

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| **Expected R&R of Research Institute** | TNO (Research Institute) expected roles and responsibilities<br><br>- Provide project coordination and technical leadership on modelling, control, and diagnostics.<br><br>- Develop and deliver physics-based models, BMS algorithms, and MEM supervisory control approaches, leveraging TNO's published and demonstrated methods [R1][R2][R4][R5][R7].<br><br>- Provide simulation and validation toolchain support (including ADVANCE) and access to relevant test facilities where applicable [R2][R3].<br><br>- Support integration, verification, and performance evaluation, including documentation suitable for transfer to embedded implementation. |
| **Expected R&R of Korean SME Partner** | Korean SME (Partner) expected roles and responsibilities<br><br>- Provide VCU/EVCU and MCU hardware and software platform, including embedded constraints, I/O interfaces, and existing baseline strategies.<br><br>- Provide system requirements, integration support, calibration data, and access to representative test environments (vehicle, bench, or HIL).<br><br>- Integrate TNO algorithms into the SME platform and contribute to verification, validation, and productization. |

## 5. Expected Outcomes/Impact

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| **Expected Outcomes and Impact** | Expected outcomes and impact<br><br>- A validated integrated software architecture for VCU/EVCU, MCU coordination, and battery monitoring that is ready for transfer into an SME controller product line.<br><br>- Improved energy efficiency and battery lifetime management via optimized energy management and charging concepts. For example, in prior work TNO demonstrated a fleet charging algorithm with 13% charging cost saving and execution time below 160 seconds for 200 vehicles in a realistic case study [R4].<br><br>- Reduced safety risk and downtime through battery and powertrain anomaly detection and diagnostics.<br><br>- Strengthened EU Korea collaboration through joint IP, demonstrators, and potential follow-up commercialization. |

## 6. References

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R8: TNO. Sustainable batteries with advanced battery management. Accessed 2025-12-17.

https://www.tno.nl/en/sustainable/mobility-logistics/batteries/

R9: Eindhoven University of Technology (TU/e). Steven Wilkins profile (notes dual affiliation TU/e and TNO and focus on battery management systems). Accessed 2025-12-17. https://research.tue.nl/en/persons/steven-wilkins/

## ☐ Terms:

**1) Industrial Technology Classification**

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

**2) Technology Readiness Level (TRL)**

| Description | | |
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| ·A consistent and objective indicator of the technological maturity of key element technologies | | |
| | **Basic Research Stage (1-2)** | 1st Stage: Basic theories and experiments<br>2nd Stage: Proof of Concept, such as practical ideas or patents |
| | **Experimental Stage (3-4)** | 3rd Stage: Verification of basic performance at a laboratory scale<br>4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale |
| | **Prototype Stage (5-6)** | 5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems<br>6th Stage: Production and performance evaluation of prototypes at a pilot scale |
| | **Productization Stage (7-8)** | 7th Stage: Reliability testing and evaluation by potential customers<br>8th Stage: Prototype certification and standardization |
| | **Commercialization Stage (9)** | 9th Stage: Commercialization |